

M. ABRAR PATEL

AI/ML Engineer | Python · LLMs & Agentic AI · RAG · PyTorch · TensorFlow

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PROFILE SUMMARY

AI/ML engineer who builds LLM-powered systems and ships them inside real products. Six months of applied ML engineering — data preparation, feature engineering, training and evaluation across NLP and computer vision in TensorFlow/Keras, plus PyTorch fine-tuning of HuggingFace transformers — alongside RAG work: embedding an internal knowledge base, indexing it in FAISS and ChromaDB, and serving semantic retrieval behind a Python API. Shipped a LangChain tool-calling agent that reads and writes live business data in a merchant platform, and build evaluation harnesses that measure whether model output is actually correct. Strong Python and CS fundamentals.

CORE SKILLS

Generative AI & LLMs: LangChain, tool calling / function calling, multi-step agent workflows, prompt design and iteration, output evaluation and reliability hardening, OpenAI-compatible APIs, chatbot design

RAG & Retrieval: document chunking, OpenAI and Gemini embeddings, semantic search, vector databases — FAISS, ChromaDB, Pinecone, pgvector

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, HuggingFace Transformers, Scikit-learn, Pandas, NumPy — neural networks, CNNs, sequence models, transfer learning and fine-tuning, feature engineering, model training, evaluation and experimentation

Languages & CS: Python (OOP, data structures & algorithms), SQL, JavaScript, TypeScript, C, C++, C# — DSA, OOP, DBMS, operating-system fundamentals

Engineering & Deployment: Django REST Framework, Flask, FastAPI, PostgreSQL, MySQL, MongoDB, Celery + Redis, Docker, Git, GitHub Actions, PyTest, Google Cloud Platform

PROFESSIONAL EXPERIENCE

AI/ML Engineering Intern — ABis Infotech Solutions

Remote · 6 months

- Built end-to-end Python ML pipelines in TensorFlow and Keras — data ingestion, preprocessing, feature preparation, model training, evaluation and inference — and fine-tuned pretrained HuggingFace transformer models in PyTorch for downstream tasks.
- Built a RAG pipeline for internal knowledge search: chunked and embedded source documents (OpenAI and Gemini embedding APIs), indexed them in FAISS and ChromaDB, and served semantic retrieval with LLM-generated answers grounded in the retrieved context — tuning chunk size, top-k and prompt structure to cut irrelevant retrievals and hallucinated answers.
- Delivered NLP and computer-vision models — next-word prediction, image captioning, face recognition, mask detection and stock prediction — each evaluated against held-out data before shipping.

Software Developer — Aarmarks Media

[aarmarksmedia.com](#) · [case study](#)

Aug 2026 – Present · TriMarkity suite, incl. the AITremarkIQ AI product

- Ship features across a live multi-app SaaS suite in Python/Django, including the AITremarkIQ AI product — collaborating with product to turn business requirements into working features.

AI PROJECTS

DukaanAI — LangChain Agent for Retail Business Data

[case study](#) · [github](#)

LangChain · Python · Flask · MySQL · SQLAlchemy · Redis · Flutter

- Built a Hindi-language LangChain agent with tool calling: the LLM selects and invokes typed tools that read and write real shop data — inventory, billing, credit — against a Flask + MySQL backend of 8 tables and 45+ REST endpoints, then composes an answer from the tool results.
- Iterated on prompts and tool schemas to make the agent reliable on noisy, code-mixed Hindi input — constraining tool arguments and adding guard rails so the model asks for missing fields instead of inventing them.
- Designed an ML-driven insights layer — dead-stock detection and demand prediction over transaction history — surfaced directly to shop owners as actions, not charts.

ModelReceipt — Verifiable LLM Inference Gateway

[modelreceipt.vercel.app](#) · [github](#) · [case study](#)

Hackathon demo — simulated attestation, capped at L0 · Next.js 15 · TypeScript · Neon Postgres · Cool SDK · Python client

- An OpenAI-compatible drop-in proxy returning a signed receipt for every completion — naming the model, version and deployment, and committing to the prompt and answer without storing either.
- Built an evaluation harness over the gateway: a 20-case golden set with deterministic graders — JSON-shape, numeric, regex and constraint checks — plus an LLM-as-judge tier deliberately held to a lower trust level and scored separately. Each case execution is sealed into a receipt, so every score is independently verifiable against the log rather than taken on the runner's word.

Coding-Task Evaluation Harness

Python · Docker · pytest · GitHub Actions

- Built a Docker-based eval harness for code-generation tasks: a Python runner takes a task folder, executes it in an isolated container, runs a hidden pytest suite against the candidate solution and emits a pass/fail JSON report — 20 task environments, 200+ hidden tests, 85% coverage, CI on every push completing in under five minutes.
- Designed tasks around the ways a model fails rather than succeeds — hidden tests the solution cannot read, catching plausible-looking answers that do not satisfy the specification.

EDUCATION

Bachelor of Computer Applications (BCA), 2023 – 2026 (final year) — Vivekanand College, Kolhapur · Prepverse ML Contest — strong overall performance